Final Report

Project Module
Mining opinions and arguments

English proficiency level prediction using SenseFlow and Graph Neural Networks

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Abstract

The study aims to clarify the relationship between the rhetorical structure of texts written by an English as a second language (L2) speaker and their L2 proficiency, as determined by an expert grade. We developed a range of predictive models for English writing proficiency, exploring different approaches that incorporate rhetorical structure features. As a supporting feature, we incorporated the demographic information about the L2 learners into a Population-based GNN. Our findings show that, while these models do not achieve state-of-the-art results, rhetorical structure proves to be an important factor in overall prediction, as confirmed by our ablation study. A graph-based approach, on the contrary, proved to be less meaningful than we presumed.

1 Introduction

It has been argued that of the four skills related to language learning, writing is far more challenging than speaking, listening and reading (Nunan, 1989). This is because the writer has to control many variables at once, including, but not limited to vocabulary, punctuation, spelling, and coherence. Consequently, second language acquisition, as well as pedagogy and linguistics, seek to study this phenomenon. Methods include the study of individual differences, case studies, the creation of written speech corpora, and contemporary experimental research (Sekerina and Fedorova, 2019).

Within this research context, argumentative essays are frequently used as a standard format for assessing writing ability, particularly in large-scale standardized examinations such as IELTS and TOEFL, as they require learners to demonstrate organization, coherence, and the ability to construct and support arguments. It has been shown that L2 writing differs from L1 writing on multiple levels, including morphology, syntax, and discourse (Hinkel, 2011). In argumentative texts written by non-native speakers, the argumentative structure is often simplified (for instance, through the increased use of basic connectors such as *and*) and implies different rhetorical strategies, often borrowed from L1.

However, the presence of simplified rhetorical structures does not necessarily imply low language proficiency. Rather, rhetorical coherence interacts with linguistic sophistication, contributing to overall writing ability. This raises the question of whether rhetorical structure, in combination with other linguistic features, can serve as a useful signal for assessing writing proficiency among L2 learners.

Our study aims to answer the following **research questions**:

1. To what extent can the rhetorical style of an L2 English learner alone contribute to predicting language proficiency (as measured by the CEFR)?

2. To what extent does incorporating learner metadata into a Graph Neural Network enhance predictive performance?

2 Related work

2.1 Second Language Acquisition

Second language acquisition (SLA) research and the linguistic studies of L2 production (such as L2 learner texts analysis) are strongly interrelated, as insights into acquisition processes rely heavily on the systematic examination of learner language.

Key areas of focus in SLA include sociocognitive perspectives, contexts of L2 use, and the influence of L1 on L2. This work serves two main purposes: to emphasize the importance of raising L2 learners' awareness of the differences between L1 and L2, and to address the question of why SLA differs so significantly from first language acquisition, particularly in case of adults (see (Pine, 2005)

and (White, 1990) for contrasting approaches to this question).

Research findings on discourse properties of L2 writing include the use of discourse particles and complex constructions, given/new (theme/rheme) idea development and argumentation planning. It is known that L2 writers generally produce shorter and less elaborative texts, appeal to personal opinion rather than proven claims and facts and utilize discourse moves differently or inconsistently due to L1 influence (Hinkel, 2011). At the same time, studies indicate that L2 learners are aware of rhetorical structure differences between their L1 and L2 (Wei and Zhang, 2020).

2.2 L2 Writing Proficiency Prediction

A variety of predictors have been used to model second language proficiency. The features listed below represent only a small subset:

- Conventinal language test results (general (MLAT[1]) as well as one-language targeted (PPVT-R) [2] (Sparks et al., 1997)
- Available demographics (age, gender, level of education) (Lett Jr. and O'Mara, 1990)
- Experimental data (e.g., neural signatures from the EEG-based research of (Ihara et al., 2021))
- Linguistic features (lexical and syntactic complexity, referential links within the text, the use of connectives) (Crossley and McNamara, 2012)

A surprising finding of the last study is that text cohesion, defined as the use of textual indicators and connectors, is not a strong predictor of L2 writing proficiency. L2 writers judged to be more advanced produce texts with fewer cohesive devices. This, however, does not align with the previous research on that matter (Jin, 2001).

All abovementioned studies rely primarily on traditional statistical methods, such as ANOVA and correlation analysis. In contrast, NLP-based approaches have only rarely addressed L2 the rhetorical/argument structure as a proficiency predictor in L2 writing. The work most closely related to ours includes (Ding et al., 2024) on transfer learning of argument mining in student essays, (Cong, 2025) on automatic essay scoring using LLMs and (Shin and Gierl, 2021) on the same task tackled with CNNs.

[1] Modern Language Aptitude Test

[2] Peabody Picture Vocabulary Test

2.3 Shallow Discourse Parsing

Shallow discourse parsing is a sub-area of discourse parsing — the identification of discourse relations between text spans. It assumes a flat hierarchy of discourse relations, with the most commonly used relations set defined in the Penn Discourse Treebank (PDTB) (Prasad et al., 2018).

The main purpose of PDTB-style annotation is to identify two (mostly consecutive) arguments *Arg1* and *Arg2*, which are connected with a *relation* (also named *sense*). Thus, we expect the input to be a text and the output to be a set of triples of the form (*Arg1, Arg2, sense*). The PDTB sense labels are listed in (Prasad et al., 2008) (v2.0). A relation can be **explicit** (if an explicit discourse connective or phrase is used) or **implicit** (if not realized explicitly).

Shallow discourse parsing enables us to model the rhetorical style and argumentation strategy of a writer — that is, how arguments are structured and presented to achieve persuasive effects in argumentative texts, and whether and how different linguistic expressions are employed (Al-Khatib et al., 2017). Prior work has approached this problem from different perspectives. (Schaefer et al., 2023) perform a fine-grained analysis of argumentation strategies by distinguishing semantic types of claims and premises, thereby modeling how arguments are constructed at the level of individual components. (Knaebel and Stede, 2023), on the contrary, model the rhetorical style of a document as a linear sequence of discourse relations and extend this approach to multiple genres and domains.

For our work, we use the **discopy** (Knaebel, 2021) parser, as it provides a general-purpose representation that does not rely on additional linguistic features (allowing us to incorporate them later if needed). The parser produces PDTB-style discourse annotations, along with POS tags and document-level positional metadata. *Discopy* is a modular parser that operates on tokenized input with contextualized word embeddings and proceeds through stages including explicit connectives and relation extraction as well as implicit sense classification. These components are modular and can be plugged in independently. At the time of its publication, *discopy* performed competitively with other state-of-the-art discourse parsers.

2.4 Graph Neural Networks (GNNs)

Graph Neural Network is a model class that allows input data to be represented using a graph structure, instead of the canonical vectorized one. This model class is often used in tasks in which information is contained not only in the data itself but also in its structure (Khemani et al., 2024). In these tasks, the use of a vector-representation-based model may lead to loss of potentially learnable information and possible misrepresentation of training and test data. Nowadays, Graph Neural Networks are widely utilized in various academic fields, especially in medicine-based ones, in which the introduction of this model class has allowed researchers to compute more faithful representations of chemical and biological structures, being able to numerically embed connections between atoms or cells (Fang et al., 2025; Paul et al., 2024).

Graph Neural Networks are becoming popular in the Natural Language Processing field as well. This is because a piece of text perfectly represents a connected structure that can be embedded as a graph, in which words are nodes and their possible connections are edges (Wu et al., 2022). All the approaches previously mentioned fall into the "Individual-Based Graph Neural Networks" subclass as they use a graph to represent each single data instance.

A Graph Neural Network can be designed to represent not only a single data instance, but also a population, creating just one graph and assigning each node to one subject (Bintsi et al., 2023). In this case, connections between nodes do not represent internal connections in a single instance of data, but rather connections among different data instances. In this models subclass, graph edges are determined by computing a similarity function between users.

Approaches that use a single graph to represent connections among different data instances are called "Population-Based Graph Neural Networks". This approach, widely used in diagnostic medicine, is particularly useful when computing prediction. The principle on which this approach is built on is the idea that similar nodes, based on the specific similarity function considered, will share similarities not only in known attributes, but also in unknown ones. This is the approach that we decided to use in our project.

When a population graph is built, three elements are generated.

1. A set of nodes N.
2. A set of edges E, representing connections between nodes.
3. A set of nodes features X, consisting of a feature matrix for each node.

N and E are used to build the adjancency matrix A, which represents the edge connections between nodes. Edge connections can also be weighted, usually ranging 0 and 1.

The general pipeline for Graph Neural Layer is the following. Given a node v and its representation h_v, the output from the Graph Neural Layer h'_v is computed as follows:

1. A function f **aggregates** $\{h_u | u \in N(v)\}$ where $N(v)$ is the neighbours set of u.
2. A function g **concatenates** h_v to the output of step 1).
3. The output of step 2) is multiplied by the weight matrix W of the graph layer.
4. A non linear function σ is applied.

Functions f, g, σ and the shape of W vary across different graph neural network architectures.

3 Data

3.1 CELVA.Sp

To ensure the diversity of the dataset and its specificity to our research interest, our models are trained and evaluated on CELVA.Sp, a corpus that includes 986 English essays written by French university students from 2018 to 2023 (Mallart et al., 2023). Participants came from two universities where English was their second language. Students were given one of three questions that they would respond to with their knowledge in their academic domains. The types of texts are ranged from descriptive, persuasive, to a combination of both. Each text is graded by expert teachers with a CEFR Level, which is used as our target and, consequently, the final output of our models. The distribution of the CEFR Levels and the distributions of the number of words and the number of sentences per text are shown in Appendix 7.1.

In addition to the English essay written by each participant, for each subject the following attributes can be found in the dataset.

- **Categorical Attributes**
    1. L1

2. Domain of specialization
3. Reading frequency
4. Other known language
5. Reinforced writing section (binary)

- **Numerical Attributes**

1. Years studying L2
2. Stay duration (weeks)
3. Stay frequency
4. Language exposure

4 Pipeline

4.1 Generation of Sense Labels

The *discopy* parser is used to generate relations and senses of EDUs for each text. The parser outputs a list of relations found in the text, each of which contains two arguments and a sense. These senses are classified into explicit and implicit relations. The labels are then further transformed into categorical features using multi-hot encoding, since any clauses in the text might act as arguments across multiple relations.

4.2 Preprocessing

4.2.1 Subject Attributes Preprocessing

During preprocessing, categorical features have been one-hot encoded, assigning to nan values (which were found mainly in the 'Other Language' attribute) an unknown label. On the other hand, numerical attributes have been normalized using a Standard Scaler, resulting in each column having $\mu = 0$ and $\sigma = 1$.

4.2.2 Sense Labels and Word Embeddings Extraction

We hypothesize that the sequential information of the sense labels obtained from the *discopy* model also predicts the proficiency level, apart from merely the frequency of their usage. Therefore, both the preprocessed representations and the models preserve the positions of the labels partially or fully to maximize the information gained from them. Additionally, this enables interactions between the representation of the text and the sense labels.

Therefore, the word embeddings and the sense labels are combined with the encoding scheme shown in Figure 2a, inspired by Zhang et al. (2019). Firstly, the text is encoded with word embeddings from a Word2Vec Model. Then each word is labelled with the type(s) of connective that it is associated with, if any. This label also indicates whether the word is part of the first argument, the second argument, or the connective itself. Finally, both the word embeddings and the label representations are concatenated to form a sequence of word-level representations available for downstream tasks, which in our case is the prediction of the proficiency level.

4.2.3 Graph Creation

During the experiment, different variations of the population graph were generated. Differences in graphs were caused by attributes chosen to base the graph on. As it will be presented in the hyperparameter tuning section, the attributes to base the graph on were an explored hyperparameter. However, regardless of the selected attributes, the general pipeline for graph generation was consistent and was organized as follows.

Given a dataset, a set of attributes C to base the graph on and a threshold t the graph adjacency matrix $A \in M^{n \times n}$, where $A^{i,j} = w$ if subject i and subject j are connected with edge weight w, was computed following these steps

1. Initialize $A^{i,j} = 0 \, \forall i, j$
2. For all $c \in C$, if c is a **categorical** attribute, $A^{i,j} = A^{i,j} + 1$ if $c(i) = c(j)$
3. For all $c \in C$, if c is a **numerical** attribute $A^{i,j} = A^{i,j} + \frac{1}{1+d}$ where $d = |c(i) - c(j)|$
4. Values in the A matrix are averaged over the number of considered attributes, $A = \frac{A}{n}$, where $n = |C|$
5. Values in the A matrix smaller than the threshold t are set to 0

This process results in the adjacency matrix $A^{i,j}$. Nodes i, j for which $A^{i,j} \neq 0$ are considered connected with edge weight $w = A^{i,j}$. As it will be presented in the next section, not all graph-based models make use of this weights. Graph Attention Networks, for example, compute their own within the neural network layer.

4.3 Evaluated Models

During this experiment, four different model classes were evaluated. Model classes were defined based on their architecture and type of input and can be seen in Table 2 in the appendix.

4.3.1 Non-Graph-Based Models Architectures: Linear Models

Linear models were asked to compute a class prediction based on the subjects attributes alone. The models evaluated were:

- Multi Layer Perceptron
- Decision Tree Classifier
- Random Forest Classifier
- Logistic Regression

This models were chosen as they often form the baseline used to compare models. They cover a variety of simple learning algorithms and require little preprocessing and computational power to work.

4.3.2 Non-Graph-Based Models Architectures: Sequential Models

With sequential models, we intend models that take as input a sequence of senses and word embeddings. As presented in the Prepocessing section, the input of these models consists in a concatenation of Word2Vec embeddings and sense labels for each word in the text evaluated. Compared to models which take as input sense sequences evaluated on a sentence level only, the sequence length increases drastically when moving to a word level, leading to more computational power needed to train a model instance. After concatenations, the input is evaluated either by a BiLSTM or by a MultiHead Attention layer. The output of this layer is then fed to a fully connected layer to produce the class prediction. During the experiment phase, these models were trained with only the embedded words as input as well, to compare the influence of sense labels in non graph-based models.

4.3.3 Graph-Based Models Architecture: Sense-only Architectures

The two model classes that feature graph-based models present a similar general architecture, with notable differences only in the Graph-based layers. The sequential input is projected into an embedding dimension (tunable parameter) and subsequently passed through a BiLSTM layer, producing an hidden representation of the input of tunable size. The hidden representation produced at the last time step is then passed, alongside with the graph edges and weights to one or Graph-based Layer, where differences arise. Depending on the model considered, two graph-based architectures were implemented : Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT).

Graph Convolutional Networks (Chen et al., 2020) aggregate neighbour features using a fixed and normalized adjacency matrix, applying the same weight to all neighbours. Weights are taken from the graph and not automatically learned. The output dimension of this layer is a tunable parameter.

Graph Attention Networks (Veličković et al., 2018) learn neighbor weights dynamically, assigning attention weights to each of them. This leads to more adaptive learning, but also to higher computational costs.

Both Graph Layers produce an output of tunable size, which is lastly passed through a dropout layer (with tunable parameters) and through a fully connected layer to finally produce the classification output.

4.3.4 Graph-Based Models Architecture: Mixed-Input Architecture

The last model that was evaluated was a Graph-Based model receiving both senses and word embeddings as input. The model architecture can be seen as a combination of the two previously described. As it happens in the sequential models, the input consists in a concatenation of Word2Vec embeddings and sense labels for each word in the text evaluated. The resulting vector is next passed through a Bi-LSTM layer. The hidden representation produced at the last time step is passed through two GAT layers and a dropout layer placed in-between, before being fed to a fully connected layer to produce the class prediction.

4.4 Training and Hyperparameter Tuning

Non-linear models were trained using Cross Entropy Loss as a loss function. Models were trained for a fixed number of epochs and with a fixed learning rates and weight decay. Train was prematurely interrupted only if the difference in loss between two epochs was less than 10^{-4}.

Hyperparameter tuning was performed using a Bayesian search algorithm from Optuna (Akiba et al., 2019). The choice of using a Bayesian search algorithm was motivated by the vast size of the hyperparameter space, which, combined with the high computational power required to train most of the models, would have resulted impossible to perform with the resources available for the experi-

ment. The Bayesian Search evaluated 200 hyperparameter configurations for each model architecture. The search space for hyperparameters and the best parameters found is presented in Tables 3, 4, 5.

4.5 Evaluation

To measure the models' performance, we used micro F1 as a primary evaluation metric, mostly due to class imbalance:

$$Precision_{micro} = \frac{TP_{total}}{TP_{total} + FP_{total}},$$

$$Recall_{micro} = \frac{TP_{total}}{TP_{total} + FN_{total}},$$

$$F1_{micro} = \frac{2 \times Precision_{micro} \times Recall_{micro}}{Precision_{micro} + Recall_{micro}},$$

where TP_{total} is the total number of true positives across all classes, FP_{total} is the total number of false positives across all classes, and FN_{total} is the total number of false negatives across all classes.

5 Results

Model Name	Micro F1
Multi-Head Attention	0.5127
LogReg	0.4595
Random Forest	0.4444
GCN	0.4353
GAT	0.4115
Decision Tree	0.3939
BiLSTM	0.3908
MLP	0.37
Multi-Head Attention (word embeddings only)	0.3553
BiLSTM (word embeddings only)	0.3451
BiLSTM_GAT_FC	0.3435

Table 1: Evaluation results (sorted in descending order)

Table 1 shows the results of the trained models with micro F1 as the performance metric. The Multi-Head Attention model performs the best with senses and full text embeddings with a micro F1 of 0.513, followed by linear models such as Logistic Regression and Random Forest with performance of 0.460 and 0.444, respectively. Despite the inclusion of senses as the input of the Graph Neural Network models, they do not outperform the linear models. Furthermore, the mixed model with bidirectional LSTM and the Graph Neural Network with a GAT layer surprisingly perform the worst with a 0.343 micro F1. Bidirectional LSTM performs worse than Multi-Head Attention by 0.122, which is within our expectations.

The performance of sequential models dropped significantly when the sense labels were removed from the model input. The performance of the Multi-Head Attention model decreases by 0.157, while the performance of the bidirectional LSTM model decreases by 0.0457.

6 Discussion

6.1 Senses Predicts Proficiency Level Only When Modeled Properly

The results partially support our hypothesis that senses contribute directly to the determination of proficiency levels when models with sense labels are compared to those without them. While Bidirectional LSTM performs better with sense labels and text embeddings only, the linear models with subject attributes only outperform the Graph Neural Networks that include both the sense labels and the subject attributes. One possible explanation is that the signals from the senses are constrained by the model architecture. The results indicate that senses are insufficiently represented with Graph Neural Networks, and this may be due to their sequential rather than graphical nature represented in texts. Specifically, with a higher performance using Random Forest as opposed to Decision Tree, the predictions made with sense labels are benefited by more complex bagging algorithms.

On the contrary, with architecture that can properly model the sequential representation of sense labels, such as Multi-Head Attention and bidirectional LSTM, sense labels are indeed highly predictive of proficiency levels. In fact, the higher performance of the attention model with sense labels compared to other models with subject attributes implies that senses, along with text embeddings, can be a stronger indicator than language-related features. Additionally, both attention and LSTM models show a significant increase in performance when sense labels are included. Therefore, models that preserve the positions of the sense labels in texts predict language proficiency better than those that do not.

6.2 Models with Senses Benefit from Attention Mechanisms

The advantage of allowing direct access between representations of different time steps in a sequen-

tial model is indicated by better performance with Multi-Head Attention layers compared to Recurrent Neural Network layers, such as bidirectional LSTM cells.

Theoretically, attention layers allow direct interactions between the representations of all time steps, whereas the representations in an LSTM cell are only communicated by passing information as short-term and long-term memories. Despite the bidirectional LSTM being able to capture contexts in both forward and backward directions, the model might fail to retain sense information as the text lengthens. In other words, the locations and the spread of the sense labels are important for the prediction of proficiency levels. For example, connectives that occur early in a text might interact with those that can be found at the end of it. Overall, this result once again stresses the importance of positions of sense when represented by a model, other than merely the difference in the types of labels.

Previous studies on proficiency level prediction leveraged attention mechanisms to maximize interactions within texts. Contextual information from texts and cross-cultural elements can both be extracted from separate attention layers to provide links between linguistic features and different cultural settings (Zainal et al., 2024). For studies aiming at automatic essay scoring, the increase in adopting attention mechanisms over recurrent neural networks are shown in recent years (Kusuma et al., 2022). In the present study, it is clear that sentence relations as a rhetorical feature, when combined with multi-head attention, is capable of predicting proficiency level of the second language.

6.3 Class Imbalance Leads to Predictions Limitations in Graph-Based models

As described in the Data section, classes in the CELVA.Sp are heavily imbalanced. Taking this into account can help explaining the poor performance of graphs-based architectures.

During a training epoch, the model is presented with the all graph at once, processing all nodes in parallel and updating the weights by averaging the loss on the whole training set. Processing the graph in batches would require the creation of subgraphs, leading to loss of information, as connections would be cut out of the graph when nodes are split in different subgraphs. This is a widely known problem in research (Wang et al., 2021), and it often arises when the classification task spans over multiple classes.

To quantify this problem, we trained each model 10 times over different stratified data samples and compared the classes distribution in the test set against the predictions. When looking at results (plots can be found in the Appendix), it can be immediately seen how all models presented a tendency in predicting lower proficiency classes, with A1 and A2 being consistently the most predicted classes, well above the true frequency in the test data.

When it comes to graph-based models, the results confirm our hypothesis. The predictions are almost entirely limited to the A1 and A2 classes, with the least frequent classes, such as B2 and C1, never being recalled.

6.3.1 Alternative Metrics

For a better evaluation of how the class imbalance problem affected models performance the following additional metrics were devised and evaluated:

1. Root Mean Squared Error : classes were mapped to progressive integers (A1:0,A2:1,...) and Root Mean Squared Error was evaluated with the following formula

$$RMSE(y, \hat{y}) = \sqrt{\frac{1}{N}\sum_i (y_i - \hat{y}_i)^2}$$

   which represents the average prediction offset.

2. Band F1: classes were mapped to the general level they refer to (A1,A2:A,...). Micro F1 was computed on the resulting data.

By looking at the results (table results for these additional metrics can be found in the Appendix.), it can be seen how the graph-based models were among the worst performing models in terms of Band F1. When compared in terms of RMSE, the models performance improved slightly. However, this result is less significant, as the RMSE score difference between the graph-based models and the other models is smaller than the one obtained in the Band F1 evaluation, which presentend a clearer ranking with broader gaps between models.

6.4 Feature Importance in Linear Models

Considering only the user attributes as input, the linear model class allows for accessible interpretability of models classification decisions. The most influent features were computed by training and

evaluating models with single features being removed from the dataset. By comparing scores to the ones obtained by training and testing on the full dataset, it is possible to list features with the highest importance for each linear model architecture. The most important features for each model were

1. **MLP** :
   - Language_exposure
   - Domain_of_specialization_Medecine
   - Domain_of_specialization_Information Communication
2. **Decision Tree** :
   - Reinforced_section_0
   - Domain_of_specialization_Medecine
   - Reading_frequency_jamais
3. **Random Forest** :
   - Domain_of_specialization_Medecine
   - Domain_of_specialization_Information Communication
   - Domain_of_specialization_Sciences de la vie et de l'environnement
4. **Logreg** :
   - Reinforced_section_0
   - Domain_of_specialization_Informatique et electronique
   - Reading_frequency_quotidienne

Feature importance results show a consistent patterns across models, denoting academic background and language exposition as key factors. This suggests a correlation between a specialization in certain academic fields and language proficiency. This is not a surprising result, considering that nowadays english is a key requisite to take part in academic research. Moreover, factors associated with the study of languages, such as reading frequency and language exposure, represent a second cluster of important features. These features are predictably important, as they are a good measure of how much effort a student is putting into learning a language, and therefore it is plausible and realistic for a correlation between the studying habits of a person and their language proficiency to exist.

7 Conclusion

The rhetorical structure of an individual's writing proves to be a promising feature for any text-based prediction. However, it cannot be a match for the state-of-the art models. Presumably, an addition of granular linguistic features (type of claim/premise, presence of specific argumentative words etc.) can positively contribute to the proficiency prediction performance.

As for the individuals' demographic features, the population-based graph derived from them did not show strong predictive power. This suggests that the assumption that people with similar backgrounds (e.g., language exposure and academic background) produce similar texts may be incorrect.

Limitations

Our study has several limitations. Firstly, the rhetorical structure is not supposed to serve as a standalone indicator of the CEFR level. In practice, overall language proficiency is multidimensional, and it is unlikely that learners uniformly exhibit the same proficiency level (e.g., C1) across listening, reading, speaking and writing. In this study, we isolate rhetorical structure as a single variable; however, future research could address this limitation by incorporating a broader range of linguistic features.

Secondly, our approach relies on a relatively small set of text parameters. While this aligns with prior L2 writing research that has focused on a limited number of linguistic features, it may not fully capture the complexity of writing proficiency.

Finally, the essays were not controlled for text length. This parameter could have washed out the predictive strength of our features.

Ethical Considerations

From a GDPR perspective, even when using the publicly available datasets, the demographics of the writers should not be transmitted to external services without explicit coverage or an ethics approval, especially since the language learner datasets tend to be small and specific. The CELVA.Sp dataset was pseudonymized: obvious variables such as name, age, email were removed. Moreover, any personal identifiable information in the essays has been replaced with the placeholder *Alex Dupont*. Still, the abundance of coding assistants (GitHub Copilot, Claude code), while active,

can send sensitive information externally from the coding environment, regardless of whether the user is actively using the tool's suggestions at the moment.

Data Availability Statement

The data that support the findings of this study are maintained in a private repository and can be shared upon request. [3]

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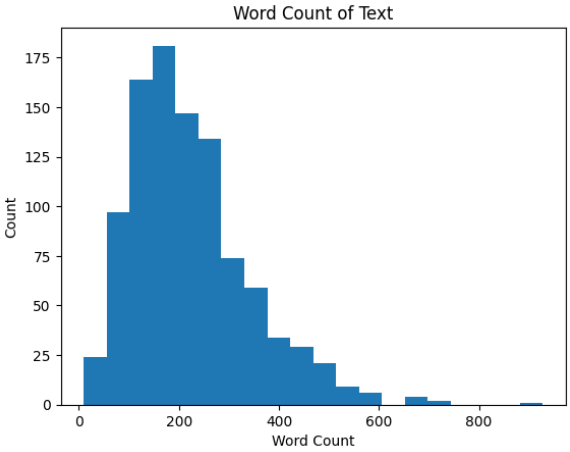
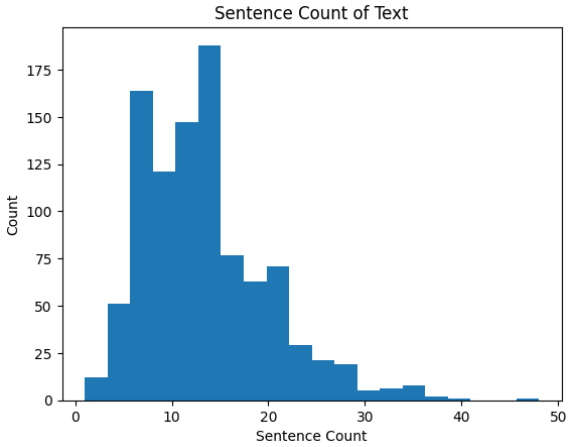
Appendix

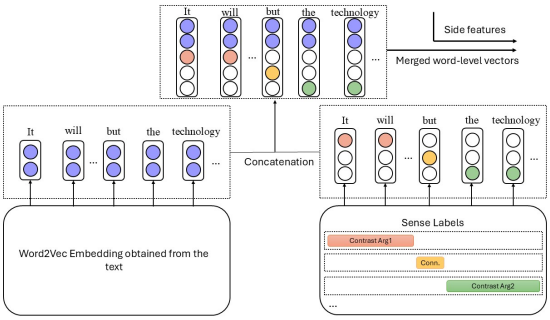
7.1 Data Preprocessing



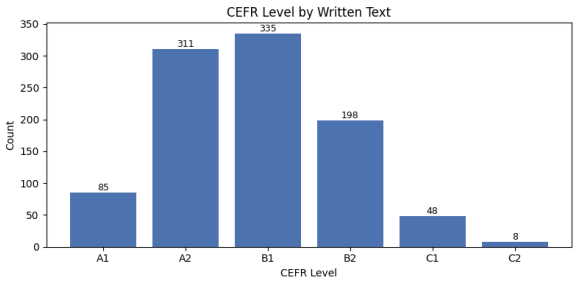
(a) Distribution of word count. The mean number of words is 224 with a standard deviation of 121.



(b) Distribution of sentence count. The mean number of sentences is 13.5 with a standard deviation of 6.47.



(a) The preprocessing procedure that combines both word and sense label representations. The resulting representation can be further combined with side information for prediction.



(b) The distribution of CEFR Levels.

7.2 Models Architecture

Model Name	Model Class	Type of Input	Model Architecture
MLP	Linear	User Attributes	Multi Layer Perceptron
Decision Tree	Linear	User Attributes	Decision Tree Classifier
Random Forest	Linear	User Attributes	Random Forest Classifier
Logreg	Linear	User Attributes	Logistic Regression Classifier
GCN	Graph-Based	Sense Sequence	BiLSTM + Graph Convolutional Layer + FC
GAT	Graph-Based	Sense Sequence	BiLSTM + Graph Attention Layer + FC
BiLSTM	Non-Graph Neural Networks with mixed input	Sense and Word Embeddings Sequence	BiLSTM + FC
Multi-Head Attention	Non-Graph Neural Networks	Sense and Word Embeddings Sequence	Multi-Head Attention + FC
BiLSTM-GAT	Graph-based with mixed input	Sense and Word Embeddings Sequence	BiLSTM + GAT + FC

Table 2: Evaluated Models Architecture

7.3 Hyperparameters Sweep

Parameter	Values
MLP	
Learning Rate	10^{-3}, 10^{-2}, $5 \cdot 10^{-3}$
Training Epochs	**50**, 100
Hidden Size	64, 128, **256**
Weight Decay	0, $5 \cdot 10^{-3}$, 10^{-2}
Activation Function	**ReLU**, Tanh
Decision Tree	
Max Depth	**3**, 5, 10, None
Min Samples Split	2, 5, **10**
Min Samples Leaf	**1**, 2, 5
Criterion	Gini, **Entropy**
Random Forest	
N Estimators	100, **500**, 1000
Max Depth	None, **10**, 20
Min Samples Split	2, **10**, 20
Min Samples Leaf	1, **2**, 5
Max Features	sqrt, **log2**, None
Logistic Regression	
C	**0.1**, 1, 5, 10
Penalty	L1, **L2**
Solver	saga, **liblinear**
Max Iter	500, 1000, **2000**

Table 3: Hyperparameters Evaluated for Linear Models (best parameters in bold)

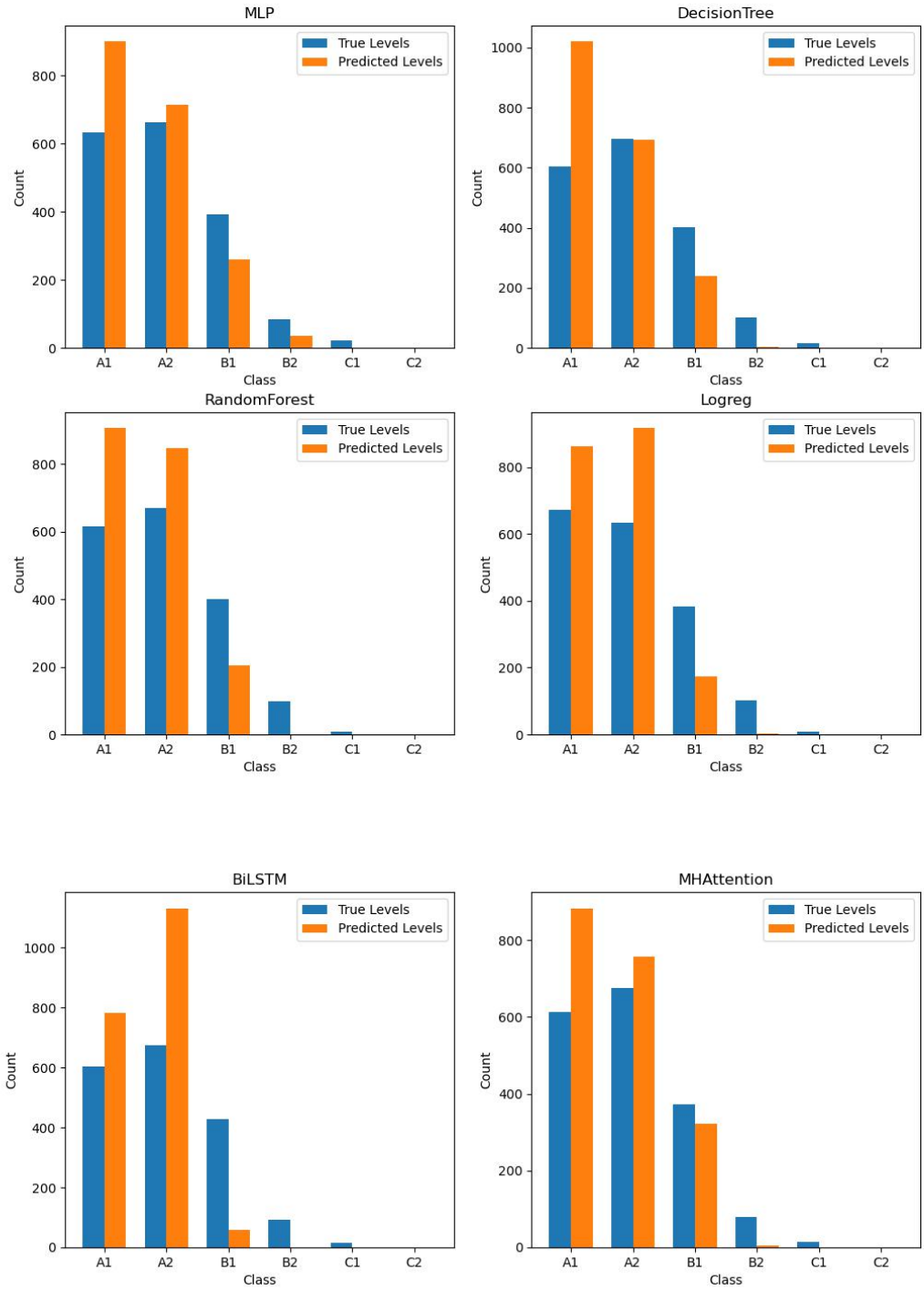
Parameter	Values
BiLSTM	
Input Size	**60**
Num Classes	**6**
LSTM Hidden Size	64, **128**, 256
Learning Rate	10^{-3}, $5 \cdot 10^{-3}$, 10^{-2}
Epochs	20, 50, **100**, 200
Weight Decay	$5 \cdot 10^{-3}$, 10^{-2}
MHAttention	
Input Size	**60**
Hidden Size	32, **64**, 128, 256
Num Heads	1, **2**, 4
Num Classes	**6**
Learning Rate	10^{-3}, $5 \cdot 10^{-3}$, 10^{-2}
Epochs	10, 20, **50**
Weight Decay	$5 \cdot 10^{-3}$, 10^{-2}

Table 4: Hyperparameters Evaluated for Sequential Models (best parameters in bold)

Parameter	Values
GCN	
GCN Hidden Size	64, **128**, 256
LSTM Hidden Size	**64**, 128, 256
Embedding Dim	16, 32, **64**
Dropout	**0**, 0.1, 0.3
Learning Rate	10^{-3}, $5 \cdot 10^{-3}$, 10^{-2}
Weight Decay	$5 \cdot 10^{-3}$, 10^{-2}
Epochs	20, 50, 100, **200**
Threshold	0.5, 0.7, **0.9**
Graph attributes	**Years, Reinforced, Exposure, Reading**
GAT	
Embedding Dim	**16**, 32, 64
LSTM Hidden Size	64, 128, **256**
GAT Hidden Size	64, **128**, 256
GAT Heads	1, 2, **4**
Dropout	**0**, 0.1, 0.3
Learning Rate	10^{-3}, $5 \cdot 10^{-3}$, 10^{-2}
Weight Decay	$5 \cdot 10^{-3}$, 10^{-2}
Epochs	20, 50, **100**
Threshold	0.5, 0.7, **0.9**
Graph attributes	Years, Reinforced, Exposure, **Reading**
GAT with Mixed input	
Input Size	**60**
LSTM Hidden Size	64, 128, **256**
GAT Hidden Size	64, 128, **256**
LSTM Layers	**1**, 2
Embedding Dim	**16**, 32, 64
GAT Heads	1, **2**, 4
Dropout	**0**, 0.1, 0.3
Num Classes	**6**
Learning Rate	10^{-3}, $5 \cdot 10^{-3}$, 10^{-2}
Weight Decay	$5 \cdot 10^{-3}$, 10^{-2}
Epochs	20, **50**, 100
Threshold	0.5, 0.7, **0.9**
Graph attributes	Years, Reinforced, **Exposure, Reading**

Table 5: Hyperparameters Evaluated for Graph Models (best parameters in bold)

7.4 Results


MLP
DecisionTree
RandomForest
Logreg
BiLSTM
MHAttention
True Levels
Predicted Levels
Count
Class
A1
A2
B1
B2
C1
C2

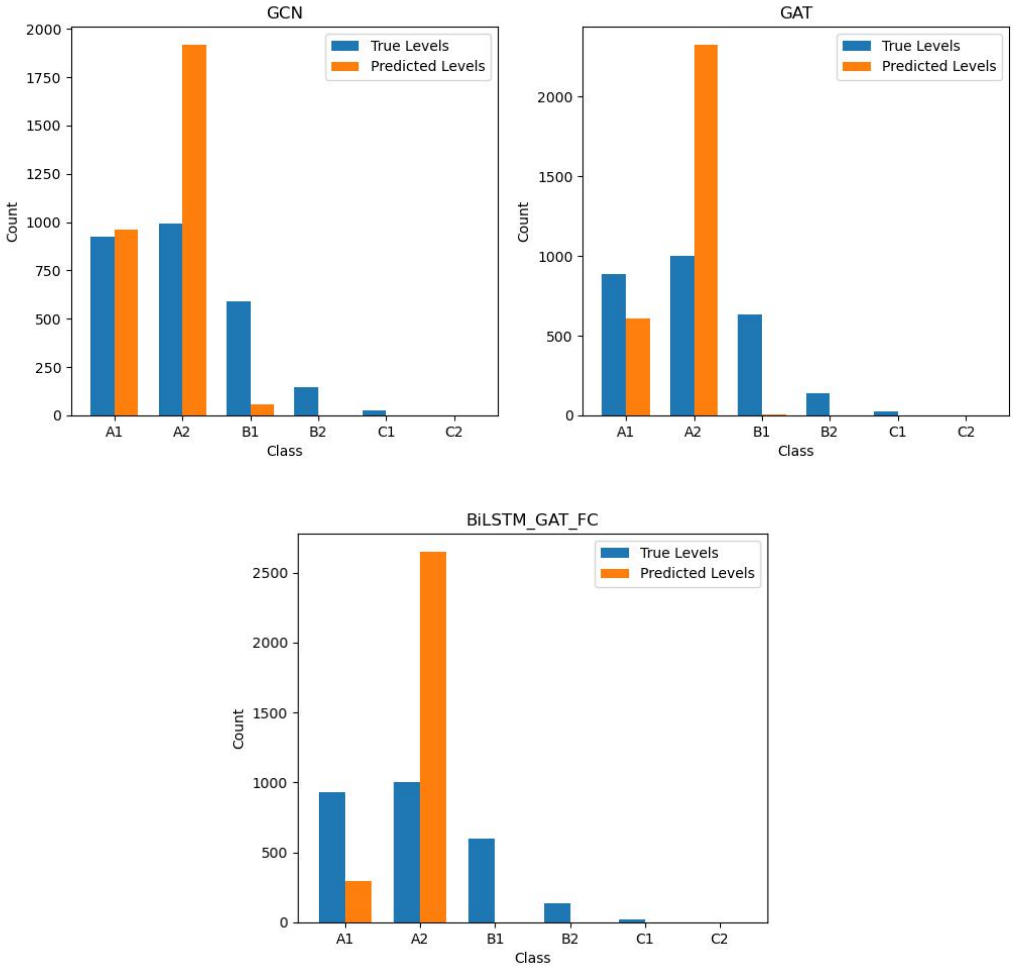


Figure 3: Comparison between real and predicted class distribution across main evaluated models